Pragati BUS MANAGEMENT SYSTEM PROJECT application

Aim

The aim of this project is to design and develop a Bus Management System using C programming language that helps in storing, managing, and updating bus details such as bus number, driver name, route, and seating capacity efficiently.

Abstract

This project focuses on the development of a simple Bus Management System using the C programming language. The application provides functionalities such as adding bus details, displaying records, searching for buses, updating information, and deleting records. The main objective of this project is to understand the concepts of structures, arrays, and menu-driven programming. By implementing this system, users can efficiently manage transportation data through a simple command-line interface. This project also improves logical thinking and problem-solving skills.

Introduction

A Bus Management System is used to organize and manage transport-related data. This project helps users store and manipulate bus details such as bus number, driver name, route, and seating capacity. It uses a menu-driven approach to perform different operations easily. This project is useful for understanding real-time applications and basic programming concepts in C language.

Software Requirements

Operating System: Windows or Linux
Programming Language: C
Compiler: GCC or Turbo C
IDE: VS Code or Dev C++

Hardware Requirements

Processor: Intel i3 or higher
RAM: Minimum 4 GB
Storage: At least 100 MB free space
Input Device: Keyboard
Output Device: Monitor

Methodology

First, the requirements of the system are identified.
Next, a menu-driven program is designed using structures in C.
Functions are implemented to add, display, search, update, and delete bus records.
Finally, the program is tested to ensure all operations work correctly.

Code

```
#include <stdio.h>
#include <string.h>

#define MAX 50

struct Bus {
    int busNo;
    char driver[50];
    char route[50];
    int seats;
};

struct Bus buses[MAX];
int count = 0;

// Add Bus
void addBus() {
    if (count >= MAX) {
        printf("Bus limit reached!\n");
        return;
    }

    printf("\n— Add Bus (Pragathi Engineering College) —\n");

    printf("Enter Bus No: ");
    scanf("%d", &buses[count].busNo);

    printf("Enter Driver Name: ");
    scanf(" %[^\n]", buses[count].driver);

    printf("Enter Route: ");
    scanf(" %[^\n]", buses[count].route);

    printf("Enter Seats: ");
    scanf("%d", &buses[count].seats);

    count++;
    printf("Bus added successfully!\n");
}
```

```
// Display Buses
void displayBus() {
    if (count == 0) {
        printf("\nNo buses available!\n");
        return;
    }

    printf("\n— Pragathi Engineering College Bus List —\n");

    for (int i = 0; i < count; i++) {
        printf("\nBus No: %d\n", buses[i].busNo);
        printf("Driver: %s\n", buses[i].driver);
        printf("Route: %s\n", buses[i].route);
        printf("Seats: %d\n", buses[i].seats);
    }
}

// Search Bus
void searchBus() {
    int no, found = 0;

    printf("\nEnter Bus No to search: ");
    scanf("%d", &no);

    for (int i = 0; i < count; i++) {
        if (buses[i].busNo == no) {
            printf("\nBus Found!\n");
            printf("Driver: %s\n", buses[i].driver);
            printf("Route: %s\n", buses[i].route);
            printf("Seats: %d\n", buses[i].seats);
            found = 1;
            break;
        }
    }

    if (!found) {
        printf("Bus not found!\n");
    }
}

// Delete Bus
void deleteBus() {
    int no, found = 0;

    printf("\nEnter Bus No to delete: ");
    scanf("%d", &no);
```

```
    for (int i = 0; i < count; i++) {
        if (buses[i].busNo == no) {
            for (int j = i; j < count - 1; j++) {
                buses[j] = buses[j + 1];
            }
            count--;
            found = 1;
            printf("Bus deleted successfully!\n");
            break;
        }
    }

    if (!found) {
        printf("Bus not found!\n");
    }
}

// Update Bus
void updateBus() {
    int no, found = 0;

    printf("\nEnter Bus No to update: ");
    scanf("%d", &no);

    for (int i = 0; i < count; i++) {
        if (buses[i].busNo == no) {
            printf("Enter new Driver Name: ");
            scanf(" %[^\n]", buses[i].driver);

            printf("Enter new Route: ");
            scanf(" %[^\n]", buses[i].route);

            printf("Enter new Seats: ");
            scanf("%d", &buses[i].seats);

            found = 1;
            printf("Bus updated successfully!\n");
            break;
        }
    }

    if (!found) {
        printf("Bus not found!\n");
    }
}

// Main Function
int main() {
```

```
    int choice;

    while (1) {
        printf("\n=== Pragathi Engineering College Bus Management ===\n");
        printf("1. Add Bus\n");
        printf("2. Display Buses\n");
        printf("3. Search Bus\n");
        printf("4. Delete Bus\n");
        printf("5. Update Bus\n");
        printf("6. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: addBus(); break;
            case 2: displayBus(); break;
            case 3: searchBus(); break;
            case 4: deleteBus(); break;
            case 5: updateBus(); break;
            case 6: return 0;
            default: printf("Invalid choice!\n");
        }
    }

    return 0;
}
```

Output

```
=== Pragathi Engineering College Bus Management ===
1. Add Bus
2. Display Buses
3. Search Bus
4. Delete Bus
5. Update Bus
6. Exit
Enter your choice: 1

--- Add Bus (Pragathi Engineering College) ---
Enter Bus No: 1
Enter Driver Name: satish
Enter Route: kakinada
Enter Seats: 50
Bus added successfully!

=== Pragathi Engineering College Bus Management ===
1. Add Bus
2. Display Buses
3. Search Bus
4. Delete Bus
5. Update Bus
6. Exit
Enter your choice: 2

--- Pragathi Engineering College Bus List ---

Bus No: 1
Driver: satish
Route: kakinada
Seats: 50
```

Advantages

Easy to use and understand
Reduces manual work
Efficient data management
Requires less memory
Improves programming and logical skills
Future Enhancements
Add graphical user interface (GUI)
Include online booking system
Connect with database
Add login authentication
Improve error handling

Conclusion

The Bus Management System developed using the C programming language successfully manages bus-related information efficiently. It helps in understanding important concepts like structures and menu-driven programming. This project provides practical knowledge of real-world applications and improves problem-solving skills.